Users & Administrators

Flash Sale Customer
- Browses flash deals
- Concurrently reserves stock
- Checks out & pays

Inventory & Sale Admin
- Schedules flash sales
- Allocates inventory quota
- Monitors live sale metrics

1. Views deals & submits buy orders (HTTPS/WSS)

Edge & Ingress Layer

CDN & DDoS Shield
(Cloudflare / CloudFront)
- Static catalog cache
- Rate limiting & bot filtering

Configures events & live inventory (HTTPS/Portal)

2. Forward validated requests (HTTPS)

Core System Boundary

SALESTORM Platform
(Central System)

- Handles 10k+ concurrent requests
- Atomic inventory reservation (TTL)
- Distributed locking & idempotency
- Order & checkout coordination

3. Authorize & verify payments (REST / Webhooks)

4. Send order dispatch manifests (REST API)

5. Trigger purchase receipts & alerts (Async Events)

External Third-Party Systems

Payment Gateway
(Stripe / Razorpay)
- Payment intent processing
- Webhook notifications

Logistics / 3PL
(Shiprocket / FedEx)
- Waybill & shipment tracking
- Order dispatch fulfillment

Notification Service
(Twilio / SendGrid / FCM)
- SMS & push receipts
- Reservation expiry alerts